注意：

允許學生個人、非營利性的圖書館或公立學校合理使用本基金會網站所提供之各項試題及其解答。**可直接下載而不須申請**。

重版、系統地複製或大量重製這些資料的任何部分，必須獲得財團法人臺北市九章數學教育基金會的授權許可。

申請此項授權請電郵 ccmp@seed.net.tw

Notice:

Individual students, nonprofit libraries, or schools are permitted to make fair use of the papers and its solutions. Republication, systematic copying, or multiple reproduction of any part of this material is permitted only under license from the Chiuchang Mathematics Foundation.

Requests for such permission should be made by e-mailing Mr. Wen-Hsien SUN ccmp@seed.net.tw

International Mathematics Assessments for Schools

2022 / 2023 JUNIOR DIVISION FIRST ROUND PAPER

Time allowed：75 minutes

When your teacher gives the signal, begin working on the problems.

INSTRUCTION AND INFORMATION

GENERAL

1. Do not open the booklet until told to do so by your teacher.
2. No calculators, slide rules, log tables, math stencils, mobile phones or other calculating aids are permitted. Scribbling paper, graph paper, ruler and compasses are permitted, but are not essential.
3. Diagrams are NOT drawn to scale. They are intended only as aids.
4. There are 20 multiple-choice questions, each with 5 choices. Choose the most reasonable answer. The last 5 questions require whole number answers between 000 and 999 inclusive. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
5. This is a mathematics assessment, not a test; do not expect to answer all questions.
6. Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are filled in. It is your responsibility that the Answer Sheet is correctly coded.

THE ANSWER SHEET

1. Use only pencils.
2. Record your answers on the reverse side of the Answer Sheet (not on the question paper) by FULLY filling in the circles which correspond to your choices.
3. Your Answer Sheet will be read by a machine. The machine will see all markings even if they are in the wrong places. So please be careful not to doodle or write anything extra on the Answer Sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

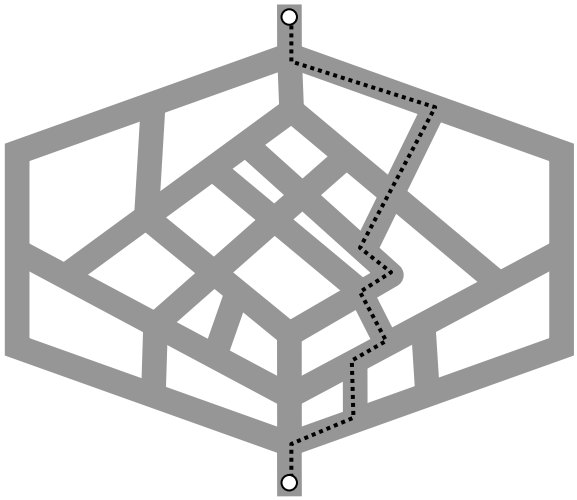
INTEGRITY OF THE COMPETITION

The IMAS reserves the right to re-examine students before deciding whether to grant official status to their scores.

2022 / 2023 JUNIOR DIVISION FIRST ROUND PAPER

Questions 1-10, 3 marks each

1. A ball is dropped starting from the top of the drainage system and slides downward until it reaches the bottom, as shown in the diagram below. If it is known that the ball cannot go upward, how many different paths could the ball pass through the drainage? (Note: One example is shown.)



（A）82　　（B）74　　（C）58　　（D）46　　（E）38

---

2. Alice buys 80 more marbles than Bob and Bob buys 67 less than Charles. Edwards buys 19 more marbles than Dennis while Charles buys 38 more than Edward. If Bob and Edward buy a total of 41 marbles between them, how many marbles are bought altogether?

（A）201　　（B）206　　（C）211　　（D）216　　（E）221

---

3. In a Mathematics test given to a certain class, the average score of those who passed is 78, while the average score of those who failed is 34. If the overall average score of the class is 67, what is the percentage of students in the class who passed the test?

（A）20%　　（B）25%　　（C）50%　　（D）75%　　（E）80%

---

4. How many integers are there between $\sqrt[3]{2022}$ and $\sqrt{2022}$?

（A）31　　（B）32　　（C）33　　（D）34　　（E）35

---

5. Two terms of an arithmetic progression are 1987 and 2022, which are not necessarily consecutive. If the common difference d of the arithmetic progression is a positive integer, then what is the number of possible values d could have?

（A）3　　（B）4　　（C）5　　（D）6　　（E）7

---

6. What is the digit sum of the numerical value of $\sqrt{11111112222222-3333333}$?

（A）0　　（B）6　　（C）18　　（D）21　　（E）24

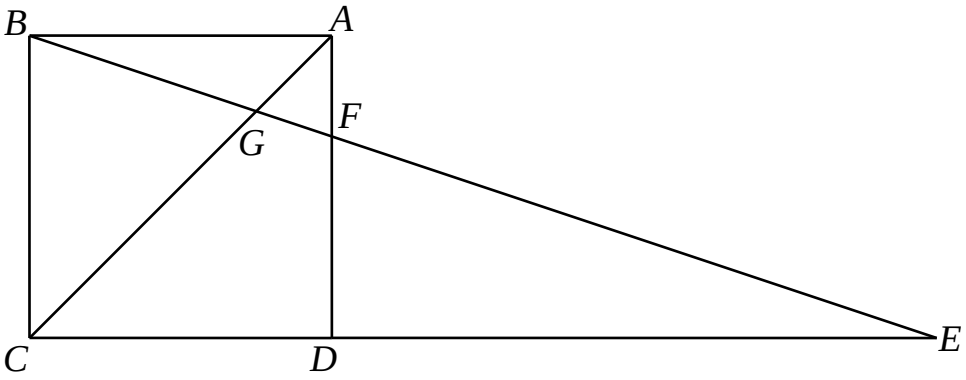
7. Which ordered pair (x, y) satisfies the following system of equations:

$$\begin{cases} \frac{1}{2}x-\frac{1}{4}y=10 \\ \frac{1}{8}x-\frac{1}{8}y=19 \end{cases} ?$$

（A）$(680, 264)$　　（B）$(-54, -87)$　　（C）$(-112, -264)$

（D）$(-112, 98)$　　（E）$(152, 264)$

8. In square *ABCD*, a line through *B* intersects the extension of *CD* at *E*, the side *AD* at *F* and the diagonal *AC* at *G*, as shown in the diagram below.



If $BG=3$ cm and $GF=1$ cm, then what is the length, in cm, of *FE*?

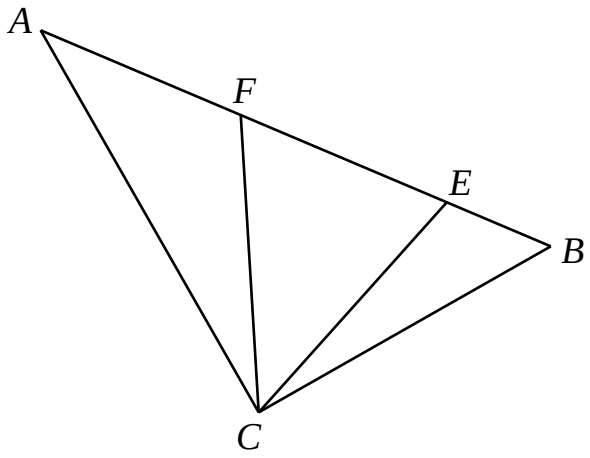
（A）5　　（B）6　　（C）7　　（D）8　　（E）9

9. The letters of the word "EVERYBODY" have been placed randomly in the grid below. By starting at the bottom left-hand corner "E", we can only move up or move to the right until we reach the top right-hand corner "Y", how many possible ways of collecting all the letters of the word in any order are there? Note that you cannot move diagonally.

	B	O	Y	R	**Y** ⇨
	D	E	E	V	D
	R	B	O	V	E
	Y	B	O	V	R
⇨	**E**	Y	V	B	O

（A）1　　（B）2　　（C）3　　（D）4　　（E）5

10. In triangle *ABC*, it is known that $\angle BCA = 90°$ and points *E* and *F* lie on the hypotenuse *AB* such that $AE = AC$ and $BF = BC$, as shown in the diagram below.



If $\angle ECF = x$, then

（A） $x = 30°$　　（B） $30° < x < 45°$　　（C） $x = 45°$

（D） $45° < x < 60°$　　（E） $x = 60°$

Questions 11-20, 4 marks each

11. A Mathematics test was given to a group of 12 students, where each problem was solved by exactly 10 students. It is known that 11 students solved exactly 7 problems each. How many problems did the 12th student solve?

（A）3　　（B）4　　（C）5　　（D）6　　（E）7

12. Two boys and five girls are to form a line. If the two boys must not be next to each other, how many different ways are there in forming the line?

（A）120　　（B）240　　（C）2880　　（D）3600　　（E）5040

13. In the base ten number system, the number 379_{10} means $3\times10^2 + 7\times10 + 9$.

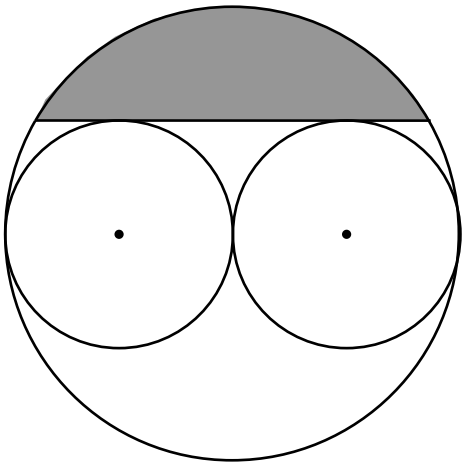
In the base *r* number system, where *r* is a positive integer, if the equation $1020_r - 460_r = 230_r$ holds, then what is the value of *r*?

（A）3　　（B）4　　（C）5　　（D）6　　（E）7

14. Alice was told the sum of three positive integers while Bob was told their product. “If I knew”, said Alice, “that your number is greater than mine, I could find the integers”. But I know that my number is less than yours.” replied Bob. What is the sum of these three numbers?

（A）4　　（B）5　　（C）6　　（D）7　　（E）8

15. The diagram below shows a head of a cartoon character which is represented by a big circle, with its eyes being two circles of radii 1 cm. The two smaller circles are touching at the centre of the big circle and they are also tangent to the big circle. The shaded region represents its hat, for which the horizontal line is tangent to the smaller circles, as shown in the diagram below.



What is the area, in cm^2, of the shaded region?

(A) $\frac{4}{3}\pi-\sqrt{3}$ (B) $\frac{4}{3}\pi$ (C) $\frac{2}{3}\pi-\sqrt{3}$ (D) $\frac{2}{3}\pi$ (E) $\pi-\sqrt{3}$

16. In the Fibonacci sequence $\{F_k\}$, $F_1=F_2=1$ and $F_{k+1}=F_k+F_{k-1}$ for $k\ge 2$. What is the value of $\frac{1}{F_1F_3}+\frac{1}{F_2F_4}+\frac{1}{F_3F_5}+\frac{1}{F_4F_6}+\frac{1}{F_5F_7}+\frac{1}{F_6F_8}+\frac{1}{F_7F_9}+\frac{1}{F_8F_{10}}$?

(A) 1 (B) $\frac{1869}{1870}$ (C) $\frac{1154}{1155}$ (D) $\frac{713}{714}$ (E) $\frac{4894}{4895}$

17. Two numbers *a* and *b* are to be selected randomly from 1, 2, 3, 4, 5, 6, 7, 8 and 9. What is the probability that $a-3b+10>0$?

(A) $\frac{35}{72}$ (B) $\frac{13}{24}$ (C) $\frac{7}{27}$ (D) $\frac{10}{27}$ (E) $\frac{13}{27}$

18. Given that $a+b+c=6$, where $1\le a$, b, $c\le 3$. What is the minimum possible value of $\frac{1}{a+b}+\frac{1}{b+c}$?

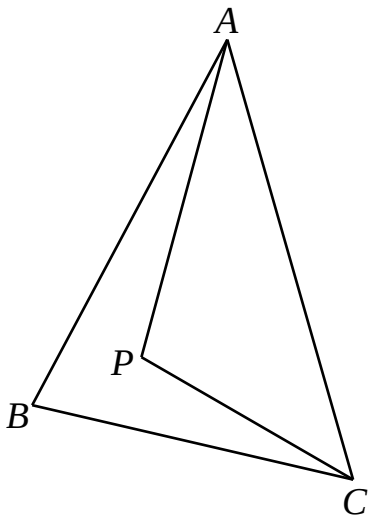
(A) $\frac{4}{9}$ (B) $\frac{9}{20}$ (C) $\frac{1}{2}$ (D) $\frac{8}{15}$ (E) $\frac{7}{12}$

19. How many different possible positive integer values of n, where $n<25$, are there such that $(n-1)^3+n^3+(n+1)^3$ is divisible by 102?

（A）1　　（B） 2　　（C）3　　（D）4　　（E）5

---

20. Let P be the point in the interior of triangle ABC such that $\angle BAP=13°$, $\angle CAP=31°$, $\angle ACP=44°$ and $AP=BC$, as shown in the diagram below. What is the measure, in degrees, of $\angle BCP$?



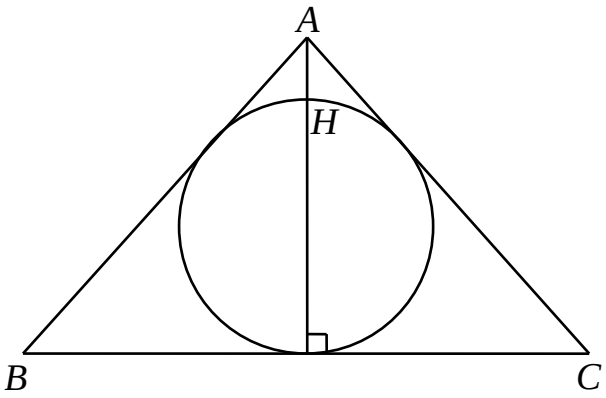
（A）13　　（B） 17　　（C）18　　（D）20　　（E）31

Questions 21-25, 6 marks each

21. Let $3^a=8(3^2+1)(3^{2^2}+1)(3^{2^3}+1)...(3^{2^6}+1)+1$. What is the value of a?

---

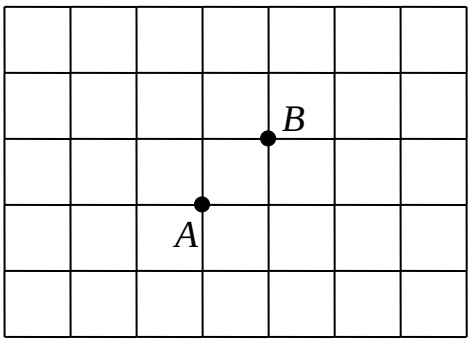
22. The orthocentre H of isosceles triangle ABC lies on its incircle, where $AB=AC$, as shown in the diagram below. If $BC=24$ cm, what is the length, in cm, of AC?



---

23. How many positive integral solutions (x, y) are there for the equation
$$1+17x+19y=xy?$$

24. The diagram shown below represents the street map of a downtown of a city. A street is a square with side length 10 m. A tourist wants to walk on the streets of this town, from his hotel (point A on the map) to the town hall (point B). Since he wants to maximize exploring the downtown, he plans to use the longest route possible, but never passing through the same point twice. What is the longest route, in m, that the tourist can traverse?



25. The unit squares on a sheet of graphing paper have side length 1. Now, a circle of radius 100 is drawn on the sheet such that it does not pass through any vertex and is not tangent to any side of any square. At most how many unit squares can the circle intersect?

***